```
manual_tfidf = tf_df * idf_df.set_index('term').T.loc['idf']
manual_tfidf = manual_tfidf.round(3)

# Display all
print(" \u2732 TF Table:\n", tabulate(round(df,3), "keys",
tablefmt="psql"))
print("\n \u2732 IDF Table:\n", tabulate(round(idf_
df.round(3),3), "keys", tablefmt="psql"))
print("\n \u2732 TF × IDF (manual recomputed):\n",
tabulate(round(manual_tfidf.round(3),3), "keys",
tablefmt="psql"))
```

The TF-IDF values for each feature word in documents 0, 1, 2, and 3 are displayed in the final TD-IDF table. While a frequent feature word (magazine) has lower scores and is therefore less significant, feature words with a rare appearance have higher TF-IDF values. These documents' key terms include 'bad', 'excellent', 'great', 'osfy', and 'quality'.

N-grams

The idea of n-grams is widely used in a variety of NLP applications, such as text prediction, information retrieval, and language modelling, where capturing the statistical characteristics of text becomes essential to understand the functionality of the underlying model. A contiguous sequence of 'n' items from a given text or speech is known as an n-gram in natural language processing. The object may be a word, syllable, character, etc. A 'unigram' is an n-gram of size 1, a 'bigram' is a size 2, and a 'trigram' is a size 3.

TF-IDF vectorization uses n-gram with an ngram_range parameter, which is a tuple (min_n, max_n). Min_n and max_n establish the lower and upper bounds of the n-gram sizes, allowing n-grams to be explored within this range. For example, if ngram_range is set to (1, 3), the 'vectorizer' will extract unigrams, bigrams, and trigrams, adding a wider context to the feature set used for machine learning models.

✲ TF Table:

	bad	excellent	great	magazine	osfy	quality
0	0	0	0.834	0.552	0	0
1	0.887	0	0	0.463	0	0
2	0	0	0.573	0.379	0	0.727
3	0	0.663	0	0.346	0.663	0

✲ IDF Table:

	term	idf
0	bad	1.916
1	excellent	1.916
2	great	1.511
3	magazine	1
4	osfy	1.916
5	quality	1.916

✲ TF × IDF (manual recomputed):

	bad	excellent	great	magazine	osfy	quality
0	0	0	1.26	0.552	0	0
1	1.699	0	0	0.463	0	0
2	0	0	0.866	0.379	0	1.392
3	0	1.271	0	0.346	1.271	0

Figure 1: TF, IDF, and TF x IDF tables